

BOLTZMANN'S LEDGER: THE THERMODYNAMIC DICTIONARY OF THE WALL

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ABSTRACT. Boltzmann's constant, like c and $\hbar$, is an exact convention; what it marks is not. It marks that entropy is a dimensionless count and that temperature is a derived rate — in the algebraic setting of the program, the *modular* rate: $k_{\text{B}}T = \hbar/2\pi$ per unit of modular flow in proper time. Taking that seriously turns out to reorganize the framework. The master equation $Z = \pi/C$ is exactly the fused Bekenstein–Hawking–Gibbons–Hawking law $4\pi S\hat{T}^2 = 1$, with $E = TS$ as the wall's Euler identity; Branch A becomes one sentence (“the squared temperature of the world is completely monotone in e-folds”), and the branch fork becomes the third-law question (temperature floor $\hat{T}_{\infty} = \sqrt{\mu_0}/2\pi$ with finite entropy, or $T \rightarrow 0$ with $S \rightarrow \infty$). The Branch-B valuation $p(N)$ is *exactly* a Fermi–Dirac occupation — forced, not chosen, because facts are projections ($P^2 = P$) — so dark energy is the thermal broadening of a fermionic level crossing: the Schottky heat capacity of the $\{0, g\}$ pair, two e-folds wide because the KMS temperature is one. Under peak-equality the magnitude problem contracts to a theorem: $\hat{\kappa} C_c = \frac{3}{4}$ exactly, $\hat{\rho}_* C_c = \frac{3}{16}$, $\hat{E}_X(\text{peak}) = \sqrt{C_c}/4\sqrt{\pi}$ — the entire dark sector is pure numbers times one dimensionless count, the capacity at the crossing, $C_c \simeq 1.49 \times 10^{122}$. “Why so small” and “why now” fuse into a single question about a single count, whose scale matches the modular-fluctuation law $\langle \Delta K^2 \rangle = \langle K \rangle$ (established for CFT vacua on balls, conjectured gravitationally) and the Poisson scaling of Everpresent Λ . Two numerological temptations are declined on the record; one (3/4) turned out to be a theorem.

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1. WHAT THE CONSTANT MARKS

Since 2019, k_{B} is exactly 1.380649×10^{-23} J/K by definition: like c and $\hbar$, it is a translation coefficient, not a fact about nature. Its content is what it translates. S/k_{B} is a pure count (nats); $k_{\text{B}}T$ is an energy, and hence a rate: temperature is not a new dimension of the world but the price of energy in entropy, $1/T = \partial S/\partial E$. In the algebraic setting the program already inhabits, this becomes precise in a stronger way: a state's modular flow $\sigma_t = \text{Ad } \Delta^{it}$ satisfies KMS at $\beta = 1$ in modular time, so *temperature is the exchange rate between modular*

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time and proper time,

$$k_B T = \frac{\hbar}{2\pi} \times (\text{modular flow rate per unit proper time}), \quad (1.1)$$

the Connes–Rovelli thermal-time reading [10]. Every famous temperature is an instance: Unruh’s $\hbar a/2\pi c k_B$ (Bisognano–Wichmann: boosts are modular flow [9, 8]), Hawking’s $\hbar \kappa/2\pi c k_B$ [6], Gibbons–Hawking’s $\hbar H/2\pi k_B$ [7].

Hawking’s formula is the Rosetta stone of the constants question:

$$T = \frac{\hbar \kappa}{2\pi c k_B}, \quad S = \frac{k_B c^3 A}{4G\hbar}.$$

Four dimensionful constants each perform one translation ($\hbar$: modular rate $\rightarrow$ energy; c , G : geometry $\rightarrow$ rate and count; k_B : energy $\rightarrow$ temperature, count $\rightarrow$ entropy) — and the physics resides entirely in two pure numbers: the 2π (KMS periodicity of modular flow) and the $\frac{1}{4}$ (the capacity normalization). The program already *owns* the 2π : it is the Borchers commutation relation $\Delta^{it} U(s) \Delta^{-it} = U(e^{-2\pi t} s)$ of the arrow postulate [11, 1], the same 2π as in R2’s $N = 2\pi t_{\text{mod}}$. The $\frac{1}{4}$ is imported with $C = A/4\ell_P^2$ [ESTABLISHED IMPORT]; Question O7 below records the one place the framework generates its own quarter.

2. THE DICTIONARY

All identities in this section are exact and verified in the companion script. Hats denote Planck units; S is in nats ($k_B = 1$).

Entropy of the wall	$S = C = A/4\ell_P^2$
Temperature of the wall	$k_B \hat{T} = \frac{t_P H}{2\pi} = \frac{1}{2\sqrt{\pi C}}$
Energy inside (Misner–Sharp)	$\hat{E} = \frac{\hat{r}_A}{2} = \frac{1}{2} \sqrt{\frac{C}{\pi}}$
Euler identity	$E = TS \quad (\text{exact})$
Master equation	$Z = \frac{\pi}{C} \iff (2\pi \hat{T})^2 = Z \iff \boxed{4\pi S \hat{T}^2 = 1}$
Poisson reading	$Z = 1/\sqrt{\hat{V}_4}, \quad \hat{V}_4 = (c/H)^4/\ell_P^4$
Physical row	$k_B T_{\text{wall},0} = \hbar H_0/2\pi \Rightarrow T_{\text{wall},0} = 2.7 \times 10^{-30} \text{ K}$

The master equation of [1] is therefore not merely *analogous* to horizon thermodynamics: it *is* the Bekenstein–Hawking entropy and the Gibbons–Hawking temperature fused into one pure-number law, $4\pi S \hat{T}^2 = 1$, holding at every epoch on the FRW wall and at the de Sitter fixed point ($\mu_0 C_\infty = \pi \Leftrightarrow 4\pi S_\infty \hat{T}_\infty^2 = 1$). The first-law route from $\delta Q = T dS$ on the apparent horizon to the Friedmann equation is Cai–Kim [13], in Jacobson’s lineage [12]; the dictionary above is its integrated, dimensionless form.

Remark 2.1 (Black holes as the other wall type). For Schwarzschild: $S = 4\pi \hat{M}^2$, $\hat{T} = 1/8\pi \hat{M}$, Smarr $M = 2TS$, and $4\pi S \hat{T}^2 = \frac{1}{4}$. The pure-number invariant $4\pi S \hat{T}^2$ takes the value 1 on the cosmological wall and $\frac{1}{4}$ on the black-hole wall — a clean, dimensionless label distinguishing the two ways causality terminates. What the fact algebra assigns to this invariant is open. [DERIVATION; OPEN READING]

3. THE DESCENT IS A COOLING CURVE

The flowed weights $z_i(N) = \mu_i e^{-\lambda_i N}$ are a Gibbs structure with the e-fold number as inverse temperature: $\beta = N$, energies λ_i , partition function $Z(N) = \sum_i z_i$. Then, exactly:

$$\langle \lambda \rangle = -D \ln Z \quad (\text{internal energy}), \quad \text{Var}(\lambda) = D^2 \ln Z \quad (\text{heat capacity}), \quad Dq = \frac{1}{2} (\langle \dot{\lambda} \rangle - \text{Var}(\lambda)). \quad (3.1)$$

The checkpoint's no-rebound identity is calorimetry: *the universe's deceleration changes at half the rate its grade spectrum stores heat*. Two conjugate readings of the same kernel coexist and should not be conflated: the *cooling* reading ($\beta = N$ grows, fixed levels; the descent cools the wall toward the L -ground state, so Λ CDM's Λ is ground-state occupation and matter is a still-populated excited grade) and the *KMS* reading ($\beta = 1$ fixed at the modular value, levels swept linearly in modular time by the Borchers rescaling). The KMS reading is the modular-covariant one; the cooling reading is the bookkeeping that makes (3.1) literal.

In this language Branch A is one sentence:

The squared temperature of the world, $(2\pi\hat{T})^2 = Z(N)$, is a completely monotone function of e -folds.

Complete monotonicity — the Bernstein condition of [3] — is the statement that the cooling curve is monotone with all derivatives alternating: cooling without rebound, reheating, or ripple. And the branch fork is the **third-law question**: Branch A halts at the Gibbons–Hawking floor $\hat{T}_\infty = \sqrt{\mu_0}/2\pi > 0$ with finite entropy $S_\infty = \pi/\mu_0$ (type II₁); Branch B cools to absolute zero, $\hat{T} \propto 1/a \rightarrow 0$, with entropy unbounded, $S \propto a^2 \rightarrow \infty$ (type II_∞). DESI, read calorimetrically, is asking whether the universe has a temperature floor.

4. THE DEFECT IS FERMIONIC

Theorem 4.1 (Fermi–Dirac occupation). [DERIVATION] *Let P be the Branch-B projection with two-block flowed weights of grades $\{\lambda_1, \lambda_2\}$, gap $g = \lambda_2 - \lambda_1$. Then the valuation is exactly a Fermi–Dirac function of e -folds,*

$$p(N) = \frac{1}{1 + e^{-g(N-N_c)}}, \quad (4.1)$$

with N_c the equipoise. The two-valuedness is forced: $P^2 = P$ means the fact has spectrum $\{0, 1\}$ — occupation at most one — so the logistic law of B2 is not an ansatz among sigmoids but the unique statistics of a projection-valued fact. A Bose–Einstein form $1/(e^x - 1)$ is excluded by idempotency (and would diverge at the crossing).

Corollary 4.2 (Dark energy as thermal broadening / Schottky anomaly). [DERIVATION] $\rho_X \propto p(1-p) = \frac{1}{g}Dp$ is the thermal broadening function $-\partial n/\partial \varepsilon$ of the level crossing, equivalently the Schottky heat capacity of the two-level pair: $\text{Var}_{\text{pair}}(\lambda) = g^2 p(1-p)$, maximal at equipoise. The observed dark energy is the Schottky bump of the wall's two-level system as the descent sweeps the level through zero; the present epoch sits at the peak (companion fits: $Dq|_0 = 0.00 \pm 0.03$). Widths are fixed by KMS normalization ($\beta = 1$): the density bump has full width at half maximum $\frac{4}{g} \text{arccosh } \sqrt{2} = 1.76$ e -folds and the occupation swings 10%–90% in $2 \ln 9/g = 2.20$ e -folds ($g = 2$). Dark energy lasts about two e -folds because the modular temperature is one.

This answers, at derivation level, a “why” the fits alone could not: why $w_0 \approx -0.8$ rather than -0.99 — because the transition width is a KMS unit, not a tunable.

5. THE CROSSING-CAPACITY THEOREM

Theorem 5.1 (Magnitude, made exact). [DERIVATION, GIVEN B4'] *Under peak-equality ($\rho_X(N_c) = \rho_m(N_c) = \rho_*$), let $C_c \equiv C(N_c)$ be the wall capacity at the crossing. Then, exactly (radiation neglected),*

$$\hat{\kappa} C_c = \frac{3}{4}, \quad \hat{\rho}_* C_c = \frac{3}{16}, \quad \hat{E}_X(\text{peak}) = \frac{\sqrt{C_c}}{4\sqrt{\pi}}, \quad \hat{T}(N_c) = \frac{1}{2\sqrt{\pi C_c}}. \quad (5.1)$$

Proof. At the peak $\rho_{\text{tot}} = 2\rho_*$, so $Z(N_c) = \frac{8\pi}{3} \cdot 2\hat{\rho}_*$ and $C_c = \pi/Z(N_c) = 3/16\hat{\rho}_*$; the rest is substitution ($\hat{\kappa} = 4\hat{\rho}_*$; $\hat{E}_X = \hat{\rho}_* \cdot \frac{4\pi}{3} Z^{-3/2}$). $\square$

The entire dark sector is now *pure numbers times one dimensionless count*: $C_c \simeq 1.49 \times 10^{122}$, the capacity of the wall at the epoch the fermionic level crosses. The coupling is exactly three-quarters of one inverse capacity; the peak density is exactly $3/16$ of an inverse capacity; the defect’s total energy inside the horizon at peak is exactly $1/4\sqrt{\pi} \approx 0.14$ of $\sqrt{C_c}$. Consequences for the honest ledger:

- (i) “*Why so small*” and “*why now*” fuse. Previously two mysteries ($\hat{\kappa} \sim 10^{-123}$ and the epoch N_c), they are now one: the value of a single count, C_c . Under B4’ that count is fixed by (Ω_m, h) ; nothing else about the dark sector is free.
- (ii) *The scale is the fluctuation scale*. $\sqrt{C_c}$ is one modular standard deviation: for CFT vacua on causal balls the modular Hamiltonian satisfies $\langle \Delta K^2 \rangle = \langle K \rangle = S$ [ESTABLISHED, LEADING ORDER] [14, 15], conjectured to persist gravitationally (Verlinde–Zurek [16]). A source worth an $O(10^{-1})$ fraction of one modular σ is exactly critical-density scale — the same Poisson scaling, $Z = 1/\sqrt{\hat{V}_4}$, by which Everpresent Λ predicted the magnitude before 1998 [17]. O2 is hereby upgraded: from “smallness relocated” to “smallness = one modular fluctuation; open residue = the mechanism selecting the crossing count and the fermionic pair.”
- (iii) *One temptation declined, one promoted*. The fitted $\hat{\kappa}C_0 = 1.120$ is *not* $9/8$: it equals $\frac{3}{2}\Omega_m(1+z_c)^3$, an Ω_m -dependent quantity [DECLINED]. The fitted $\hat{\kappa}C_c = 0.750$, first flagged as a suspicious rational, is Theorem 5.1 [PROMOTED TO THEOREM].

6. NEW FORMAL OBJECTS

Conjecture 6.1 (C8: KMS strip). *With $N = 2\pi t_{\text{mod}}(R2)$, the KMS analyticity of the descent state should appear as analytic continuation of $Z(N)$ to the strip $|\text{Im } N| < \pi$, complete monotonicity being the boundary shadow of strip analyticity, and $\text{Im } N$ playing the role of modular phase (the Fourier dual recovering the grade spectrum).* [CONJECTURE]

Observation 6.2 (Defect poles at half depth). The defect $\text{sech}^2(N - N_c)$ continues to complex N with poles at $\text{Im}(N - N_c) = \pi/2 + \pi\mathbb{Z}$ — inside the conjectured KMS strip, at exactly half its depth. Whether this is the analytic signature of the crossing or an accident of the sech-squared form is open. [OBSERVATION]

Question 6.3 (O7: the two quarters). The framework contains two quarters: the imported capacity normalization $C = (A/\ell_P^2) \cdot \frac{1}{4}$, and the derived variance bound $\max_p p(1-p) = \frac{1}{4}$ of a binary fact. If each Planck cell of the wall carries one $\mathbb{Z}/2$ fact — the sign character the A_2 register already supplies [2] — and capacity counts *maximal binary variance per cell* (equivalently, linear entropy / purity deficit, the quantity the two-block proposition already met), then the Bekenstein–Hawking quarter and the Bernoulli quarter are the same quarter. Flagged hard: the standard it-from-bit counting gives $\ln 2$ per cell, not $\frac{1}{4}$, so this reading reinterprets what capacity counts; it is a conjecture with numerology risk, recorded because it is the framework’s only self-generated $\frac{1}{4}$. [CONJECTURE; NUMEROLOGY RISK]

Remark 6.4 (The three-temperature program: R2 sharpened). Hawking’s derivation succeeded by forcing three separately established structures — field theory, geometry, thermodynamics — to agree, and the agreement fixed the pure numbers. The program’s analog: demand that the wall’s *modular* temperature (from the HSMI/Borchers structure), its *geometric* temperature ($\hbar H/2\pi k_B$), and its *calorimetric* temperature (the KMS $\beta = 1$ of the descent Gibbs kernel) coincide. The triple match is the sharpened form of the identification conjecture R2, and it is the kind of consistency that would fix the framework’s remaining normalizations the way Hawking’s fixed the quarter. [PROGRAM]

7. MORGUE ADDITIONS

M5. $\hat{\kappa}C_0 = 9/8$: declined; the quantity is Ω_m -dependent (Section 5).

- M6.** Komar-energy factor games ($E = 2TS$ vs $E = TS$ bookkeeping): the note commits only to the Misner–Sharp identity $E = TS$, verified exactly; other energy definitions shift Smarr coefficients and are not imported.
- M7.** Varying- k_B (or varying- c , $\hbar$) readings of any of the above: nothing here varies a convention; all content is carried by dimensionless counts and pure numbers.

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